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  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:33:01",
  "variant" : 2,
  "subjectName" : "mathematics",
  "categories" : [ "Planimetry", "Probability", "Combinatorics" ],
  "questions" : [ {
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    "answers" : [ "15", "17", "14", "13" ],
    "correctIndex" : 3
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    "question" : "Which theorem states that the square of the hypotenuse is equal to the sum of squares of the other two sides?",
    "answers" : [ "Euclid's Theorem", "Ptolemy's Theorem", "Pythagorean Theorem", "Thales's Theorem" ],
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    "answers" : [ "10 * pi", "25 * pi", "50 * pi", "100 * pi" ],
    "correctIndex" : 1
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    "question" : "What does Bayes' Theorem calculate?",
    "answers" : [ "Discrete distributions", "Joint probability", "Marginal probability", "Conditional probability" ],
    "correctIndex" : 3
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    "question" : "If events A and B are independent, what is P(A and B)?",
    "answers" : [ "P(A) / P(B)", "P(A) * P(B)", "P(A) + P(B)", "P(A) + P(B) - P(A or B)" ],
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    "question" : "What is the sum of probabilities of all possible mutually exclusive outcomes in a sample space?",
    "answers" : [ "1", "Infinite", "100", "0" ],
    "correctIndex" : 0
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    "question" : "How many distinct subsets can be formed from a set containing 5 elements?",
    "answers" : [ "16", "32", "25", "64" ],
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    "question" : "What is the value of the combinatorial coefficient C(5, 2)?",
    "answers" : [ "20", "5", "15", "10" ],
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}, {  
  "question" : "How many unique permutations can be made from the letters in the word 'APP'",  
  "answers" : [ "9", "1", "6", "3" ],  
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  "question" : "Which principle is also known as the Dirichlet Box Principle?",  
  "answers" : [ "Inclusion-Exclusion Principle", "Multiplication Principle", "Pigeonhole Principle",  
    "Bijection Principle" ],  
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